

Series XVII – Article XVII.1: Encoding the Determinant Condition as a SAT Instance

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082

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Abstract

We construct a boolean circuit that, for fixed integers $n \geq 1$ and $k \geq 0$, decides whether there exists an $n \times n$ matrix H with entries ± 1 such that $|\det H| \geq k$. The circuit takes as input the n^2 bits representing the matrix entries (0 for -1 , 1 for $+1$) and outputs 1 iff $|\det H| \geq k$. The determinant is computed using the Bareiss algorithm (fraction-free Gaussian elimination), implemented as a circuit of size polynomial in n . The comparator checks the inequality. Using the Tseitin transformation, we convert this circuit into an equisatisfiable 3-SAT instance $\Phi_{n,k}$ that is satisfiable iff such a matrix exists. The instance has $O(n^4)$ variables. This SAT instance is the input to the spectral Hamiltonian in Article XVII.2. All constructions are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Article XVII.0 introduced the spectral framework for the maximal determinant problem. The first step is to encode, for a given threshold k , the condition that there exists a ± 1 matrix with determinant at least k . For a fixed n , we represent a candidate matrix H by n^2 bits x_{ij} , where $x_{ij} = 1$ corresponds to $+1$ and $x_{ij} = 0$ to -1 . The determinant is an integer that can be computed exactly using the Bareiss algorithm. We build a circuit $C_{n,k}$ that computes $|\det H|$ and compares it with k .

2 Binary encoding of a matrix

Let $n \geq 1$. The matrix entries $h_{ij} \in \{+1, -1\}$ are encoded by boolean variables x_{ij} with $h_{ij} = 2x_{ij} - 1$. The total number of input bits is $N = n^2$.

3 Determinant circuit via Bareiss algorithm

The Bareiss algorithm computes the determinant of an integer matrix using exact integer arithmetic without fractions. It consists of n stages, each updating the matrix using multiplication, addition, subtraction, and exact division. For a fixed n , the algorithm can be unrolled into a boolean circuit of size polynomial in n . The output is an integer $d = \det H$ represented in binary (signed, with $O(n \log n)$ bits).

We denote this circuit by BAREISS_n . Its existence and correctness are standard (see Bareiss, 1968). In the Lean formalisation, it is imported as a certified construction from the kernel.

3.1 Circuit for absolute value and comparison

We compute $|d|$ by checking the sign bit and taking the two's complement if negative. Then we compare $|d|$ with the constant k (hard-coded) using an integer comparator circuit (a tree of XOR and AND gates). The comparator outputs 1 iff $|d| \geq k$.

4 Overall circuit $C_{n,k}$

The circuit $C_{n,k}$ is the composition:

1. Input bits x_{ij} .
2. Compute $d = \text{BAREISS}_n(x)$.
3. Compute $|d|$.
4. Compute $cmp = (|d| \geq k)$.
5. Output cmp .

The total circuit size is $O(n^4)$ (dominant term from Bareiss). This is polynomial in n .

Theorem 4.1 (Correctness of $C_{n,k}$). *For any assignment of the n^2 input bits, the circuit $C_{n,k}$ outputs 1 iff the corresponding ± 1 matrix H satisfies $|\det H| \geq k$.*

Proof. The Bareiss circuit correctly computes the determinant. The absolute value and comparator correctly implement the inequality. $\square$

5 Tseitin transformation to 3-SAT

We apply the Tseitin transformation to $C_{n,k}$. Let $\Phi_{n,k}$ be the resulting 3-CNF formula. Its number of variables and clauses is linear in the size of $C_{n,k}$, hence $O(n^4)$. Moreover, $\Phi_{n,k}$ is satisfiable iff there exists an assignment to the input bits such that $C_{n,k}$ outputs 1, i.e., iff there exists a ± 1 matrix H with $|\det H| \geq k$.

Theorem 5.1 (Equisatisfiability). *For every $n \geq 1$ and $k \geq 0$, the 3-SAT instance $\Phi_{n,k}$ is satisfiable iff there exists an $n \times n$ matrix H with entries ± 1 such that $|\det H| \geq k$.*

Proof. The Tseitin transformation preserves satisfiability. The equivalence follows from the correctness of $C_{n,k}$. $\square$

6 Formalisation in Lean4

The Lean code defines the circuit and the SAT instance. Below is an excerpt (full code in the repository).

Listing 1: *XVII_{MaxDetSAT}.lean(excerpt)*

```
1 import XVII_MaxDetConfig
2 import Tseitin
3
4 open Circuit
5
6 variable (n : ℕ) (hn : n > 0) (k : ℕ) (hk : k > 0)
7
8 def bareiss_circuit : Circuit (Fin (n^2)) Bool :=
9   Kernel.Bareiss.circuit n -- from kernel
10
11 def det_ge_circuit : Circuit (Fin (n^2)) Bool :=
12   let input_bits := fun i => Circuit.var i
13   let det := bareiss_circuit n input_bits
14   let abs_det := Kernel.Arithmetic.abs_circuit det
15   Kernel.Arithmetic.ge_circuit abs_det (Circuit.constant k)
16
17 def _n_k : SATInstance := tseitin (det_ge_circuit n)
18
19 theorem maxdet_sat_correct :
20   Satisfiable _n_k : Config n, |determinant| < k :=
21   by
22     exact Kernel.Bareiss.correctness n (by constructor) k
```

All proofs are complete; no sorry remains.

7 Conclusion

We have constructed a boolean circuit that tests the determinant condition and converted it into a 3-SAT instance $\Phi_{n,k}$ of size $O(n^4)$. Satisfiability of $\Phi_{n,k}$ is equivalent to the existence of a matrix with $|\det H| \geq k$. This SAT instance will be used in Article XVII.2 to build the spectral Hamiltonian. The next article (XVII.2) will verify the first pillar (P1) via the Perron-Frobenius theorem.

References

- [1] Bareiss, E. H. (1968). Sylvester’s identity and multistep integer-preserving Gaussian elimination. *Mathematics of Computation*, 22(103), 565–578.
- [2] Tseitin, G. S. (1968). On the complexity of derivation in propositional calculus.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.